

Theme 3: AI in Hardware & Manufacturing

(Intelligent Machines, Edge Computing & Cognitive Production)

SDG Alignment: **SDG 8** (Decent Work and Economic Growth), **SDG 9** (Industry, Innovation, and Infrastructure)

ANRF Alignment: Advanced Manufacturing, AI & Robotics, Industry 4.0/5.0 Technologies

- **Predictive Maintenance & Machine Diagnostics**
 - Sensor-based vibration, thermal, and electrical monitoring for early fault prediction.
- **Computer Vision & Smart Quality Inspection**
 - Real-time vision hardware for defect detection, surface inspection, and automated sorting.
- **Intelligent Robotics & Collaborative Automation**
 - Adaptive robotic arms, Cobots, CNC integrations, and AI-driven pick-and-place systems.
- **Generative Design & Digital Twins**
 - AI-driven CAD hardware design for 3D printing/machining and real-time physical digital twin mirrors.
- **Edge AI & On-Chip Processing**
 - Lightweight AI microcontrollers and hardware modules for real-time factory-floor decision-making.
- **Worker Safety & Hazard Prevention Systems**
 - AI-powered smart wearables and vision-based zone monitoring to predict and stop industrial accidents.